

COSMO: An AI-Native System for B2B Sales Outreach with AI-Nativeness Auditing

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Abstract. Small and medium Business-to-Business (B2B) firms want to scale outreach without hiring proportionally, and every sales tool now advertises itself as AI-native, yet the term has no testable meaning: it is applied both to rule-based platforms that bolt on an optional assistant and to systems whose core workflows cannot run without a language model. This paper makes three contributions. First, it gives Artificial Intelligence (AI)-native an operational definition for sales and customer relationship management (CRM) systems as six checkable criteria, and audits four widely-used platforms against them, finding that none satisfied all six. Second, it presents COSMO, a working system built to satisfy every criterion, separating COSMO’s AI-native characteristics from the technical features that support them, and showing how the system serves a B2B business that already has customers. Third, because COSMO is a system and not only a model, we evaluate it across three fronts — frontend responsiveness, backend throughput, and the quality of its two most consequential AI functions — the last with an LLM-as-a-Judge protocol in which candidate and judge model families are kept disjoint and the judges are drawn from the strongest models outside the candidate families. On retrieval-grounded reply drafting COSMO scores 4.88 out of 5, marginally above Claude Sonnet 5 (4.81) and Gemini 3.1 Pro (4.82). Such results confirm the contribution of our COSMO to B2B commerce context.

Keywords: AI-native systems · large language models · agentic AI · retrieval-augmented generation · LLM-as-a-Judge · B2B sales outreach · human-in-the-loop

1 Introduction

1.1 Context and current state

For a B2B firm, reaching external decision-makers is not a single message but a multi-step process — initial contact, follow-up, scheduling, clarification — that demands consistent execution and the continuous reuse of prior context. For

small and medium enterprises this is a bottleneck: they rarely have the budget for a dedicated business-development function, contact-enrichment services, or CRM tooling. Run by hand across email threads, spreadsheets, and personal notes, outreach fragments: context is lost, follow-up depends on individual memory, and no shared, auditable record exists. The cost is measurable — one field study found that more than seventy percent of marketing-generated leads are never followed up [2] — and prior work is explicit that automation helps only when calibrated to the buyer journey and kept under human oversight, not when left to run open-loop [11]. Vendors respond with AI features and, almost universally, the label “AI-native.” The label, however, has followed the same inflation “cloud-native” once did [1]: it is attached to any product that ships a model, whether or not the model is load-bearing, so a buyer cannot tell whether a system *depends* on AI or merely *contains* it.

1.2 Objective and contributions

This gap motivates two linked goals: making AI-nativeness testable rather than merely claimed, and building a system that satisfies it in a real B2B setting. Concretely, we (i) define operational AI-nativeness for outreach and CRM systems as six checkable criteria and apply it as an audit of four platforms (Section 3); (ii) present COSMO, a system meeting that definition, separating its AI-native characteristics from the technical features that support them (Section 4); and (iii) evaluate COSMO across frontend, backend, and AI quality with a bias-hardened judging protocol (Section 5). Practically, the work is a democratization argument: by packaging capabilities that were previously enterprise-only — enrichment, disciplined sequencing, next-best-action prioritization — into one system, COSMO lets a small team run outreach at a quality that used to require a dedicated sales-development function, with the model handling repetitive, data-heavy work while relationship-building and closing stay with the human.

2 Related Work

Automation platforms. Contact-centric platforms such as HubSpot [3] and Salesforce Marketing Cloud [4] pair a record store with a workflow engine and have added email drafting and next-best-action prompts. Sales-engagement tools such as Outreach [5] add reply sentiment classification over cadences, and open-source suites such as Mautic [6] provide self-hosted automation with no generative capability. In all of these works, the automation is authored in advance; the model, where present, decorates a control flow it does not direct.

Large Language Model (LLM) agents in the enterprise. A newer line treats the model as the controller. CRMArena [7] benchmarks language agents on realistic CRM tasks and finds that even capable models struggle with procedural discipline — a caution we carry into our design. We adopt Anthropic’s distinction [8] between a workflow whose tool sequence a developer fixes and an agent that chooses tools at run time, consistent with the classical view of an agent.

AI in sales and evaluation. Paschen et al. [9] argue human and machine intelligence are complementary along the funnel; Cao and Zhu [10] formalize next-best-action as learning. Retrieval-augmented generation [12] grounds output and lowers hallucination in structured tasks. Since we evaluate with model judges, we also rely on the LLM-as-a-Judge literature: the method tracks human preference [13] but is biased [14, 15].

3 The AI-Native Criteria

Because the term is used loosely, we first state what we take an AI-native outreach system to be, then apply it. Drawing on collaborative intelligence [9], next-best-action learning [10], agentic systems [8], and the oversight automated nurturing requires [11], we define six criteria, each phrased so its presence can be checked directly.

C1: Critical-path dependence — disabling the model breaks core workflows rather than merely slowing them. *C2: Agentic mediation* — a model selects and composes backend operations from a registry at run time, not from a developer-fixed script. *C3: Retrieval grounding* — generated text is conditioned on retrieved, inspectable enterprise evidence, so an output can be checked against its sources. *C4: Feedback capture* — human accept/revise/reject decisions are persisted as first-class signals. *C5: Learned intent routing* — a learned classifier, not keyword rules, maps inbound replies to downstream actions. *C6: Supervised autonomy* — no irreversible external action executes without a human check-point.

The six were synthesized from the cited literature; whether they actually discriminate is checked directly by Table 1, which audits four platforms against them from each platform’s own public documentation. A system missing even one is, on this reading, a rule-based product with AI attached rather than an AI-native one: the commercial tools earn partial marks because their AI is real but optional, the open-source suite earns none, and none clears every criterion.

4 COSMO: System Design

4.1 Overview and whom it serves

COSMO is built for small and medium firms that sell to *other businesses* — SaaS vendors, agencies, consultancies — and scopes to the pre-sale lead pipeline, handing the deal itself back to a human. Prospects enter (identify) via Apollo.io, CSV import, a LinkedIn extension, or a lead form; are enriched (research) with AI-inferred pain points and buying signals scored against an ideal-customer profile; receive a personalized first message (outreach); are followed up on an adaptive cadence (nurture); and are handed to a human with a pre-meeting briefing (engage). Each prospect carries an explicit outreach state — cold, no-reply, replied, post-meeting, dropped — advanced automatically from email events and classified intent, giving the shared, auditable record manual outreach lacks.

Table 1. Platforms Audited Against the Criteria (● met, ○ partial, — absent)

Criterion	HubSpot[3]	Salesforce[4]	Outreach[5]	Mautic[6]	COSMO
C1: Critical-path dependence	○	○	○	—	●
C2: Agentic mediation	—	—	—	—	●
C3: Retrieval grounding	○	○	—	—	●
C4: Feedback capture	○	○	○	—	●
C5: Learned intent routing	—	○	○	—	●
C6: Supervised autonomy	○	○	●	—	●

4.2 AI-native characteristics of COSMO

We separate *what makes COSMO AI-native* (this subsection, mapped to the criteria) from *the technical features that support it* (next subsection). *C1: Critical-path dependence.* Template generation, reply drafting, enrichment, and the daily briefing all call the model on the request path; removing it halts these workflows rather than slowing them, since none has a rule-based fallback. *C2: Agentic mediation.* A conversational business-development agent takes a natural-language request and fulfils it by composing calls over roughly sixty backend operations chosen at run time, rather than a coded sequence. *C3: Retrieval grounding.* Every generation prompt is assembled with retrieved knowledge-base chunks and structured contact fields, so an answer about pricing or features can be checked against real records. *C4: Feedback capture.* The system records whether each AI draft was used, edited, or discarded, and whether each insight was confirmed or rejected, into dedicated feedback tables. *C5: Learned intent routing.* A learned classifier maps every inbound reply to one of nine intents (with an explicit *other* fallback to a human) and dispatches it to a handler, as Table 2 details. *C6: Supervised autonomy.* Every outbound message passes a human checkpoint at the level appropriate to the action: an AI-drafted reply is held for per-message approval, and a campaign or playbook sends on schedule only after a person composes, reviews, and activates it. The write the system performs on its own is suppression — flagging a contact who opts out — which only ever reduces outbound activity.

4.3 Technical features supporting the design

The backend is a modular monolith — a single Go service (Fiber, GORM) in a handler–service–repository layering — beside a Next.js frontend and asyn-

Table 2. Inbound Intent Labels and Routed Actions

Intent	Routed action
INTERESTED	draft a reply immediately
REQUEST_FOR_PRICING	attach pricing context; draft reply
REQUEST_FOR_INFORMATION	attach product context; draft reply
REFERRAL	create a new lead; draft reply
NURTURE	continue the nurture cadence
NOT_INTERESTED	stop the sequence
OUT_OF_OFFICE	pause and reschedule
DO_NOT_CONTACT	suppress the contact
OTHER	no automated action; assign to a human

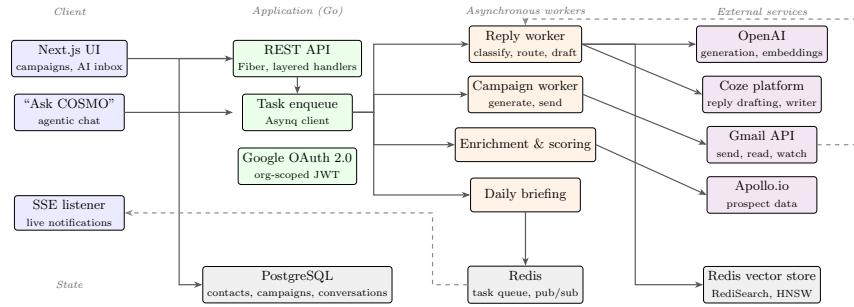


Fig. 1. COSMO system architecture. Solid arrows are synchronous calls; dashed arrows carry inbound Gmail push notifications into the reply worker and realtime events back to the browser. Persistent state sits on its own row: every model call runs on a worker, so the request path stays responsive.

chronous workers. Because each AI call takes several seconds, generation, enrichment, and reply processing run on workers via a Redis-backed queue, keeping the request path responsive (supporting C1). PostgreSQL is the system of records with JSONB for evolving fields. Redis serves three roles — task queue, cache, and vector store via Redisearch with HNSW indexing — powering semantic retrieval over contacts and knowledge (supporting C3). Generation uses an OpenAI GPT model and retrieval `text-embedding-3-small` at 1536 dimensions. Real-time notifications reach the browser over SSE fanned out through Redis pub/sub (supporting C6’s review loop). Authentication is Google OAuth with organization-scoped JWTs. Figure 1 shows how these parts fit together, and Table 3 summarizes the stack.

4.4 The autonomous reply pipeline

The tightest illustration of the design is the reply pipeline, shown in Figure 2. A push notification enqueues a job; a worker fetches the message, links it to a contact and campaign, and stores it. The model classifies the reply intent; `DO_NOT_CONTACT` blacklists the contact and `OUT_OF_OFFICE` pauses the cadence, while any actionable intent triggers retrieval and a grounded draft. The draft

Table 3. COSMO Technology Stack

Layer	Technology
Frontend	Next.js, React, TypeScript
Backend	Go, Fiber, GORM
Database	PostgreSQL (JSONB)
Queue / cache	Redis + Async
Vector store	Redis RediSearch (HNSW)
LLM	OpenAI GPT (generation), <code>text-embedding-3-small</code>
Object storage	AWS S3
Auth	Google OAuth 2.0, JWT

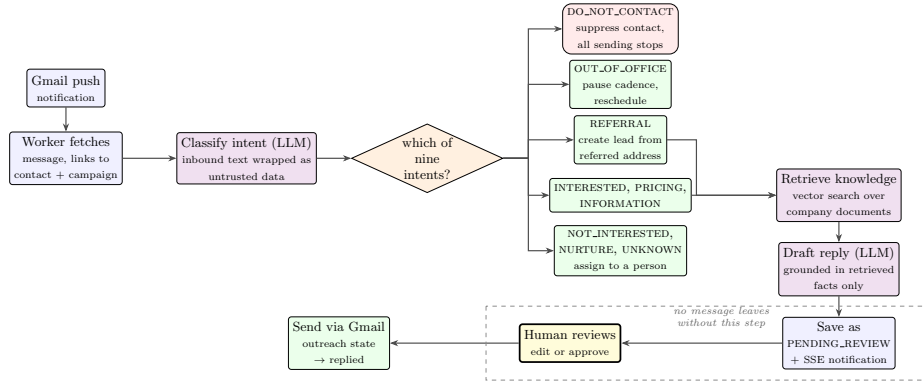


Fig. 2. The autonomous reply pipeline. Everything up to the dashed region runs without human intervention: classification, routing, retrieval and drafting. The draft is then held for review, and only a person can release it — autonomy in execution, human control at the point where a message would leave the system. Opt-out is the one branch that never produces an outbound message at all.

is saved, the outreach state moves from *no-reply* to *replied*, and an SSE event surfaces a notification. The chain runs without human intervention up to that point — autonomy in execution, human control at the place it matters, before anything is sent (C5, C6).

4.5 Data handling and responsible use

We limit processing to the outreach features a customer signs up for and do not sell, rent, or use customer data to train any shared model. In the reply pipeline, inbound emails and retrieved documents are treated as untrusted data rather than instructions: they sit in delimited context blocks noting their contents are data, not commands; classifier output is validated against the fixed intent set; and no AI-drafted reply reaches a recipient without human approval, so an injected instruction cannot cause a message to be sent. We honor opt-out — a `DO_NOT_CONTACT` reply flags the contact once processed, after which every auto-

mated send path skips it — and operators remain responsible for the anti-spam and data-protection rules of their jurisdiction.

5 Evaluation

Because COSMO is a system, we evaluate three fronts: frontend responsiveness, backend throughput, and the quality of the AI function that most directly affects external communication.

5.1 Frontend performance

We ran Google Lighthouse (desktop) on three representative pages: the login page (/), the AI inbox (/ai-inboxes), and the outreach page (/outreach). Table 4 reports category scores. SEO is perfect throughout and Best Practices is perfect on the authenticated pages; the login page is lower only because the test ran over HTTP on localhost. Performance is the weakest axis, driven by a large shared JavaScript bundle (high total blocking time) and an unoptimized hero image (high LCP on the login page): first paint arrived at 1.4s but the remaining forty seconds of the login page’s pre-fix LCP (41.3s) were spent fetching one 3.78 MB hero PNG.

Both diagnoses were then acted on, and both proved correct. Converting the landing artwork to WebP at the width it renders reduced those assets from 15.2 MB to 3.0 MB, the hero alone to 380 kB. For the bundle, the largest single contributor was not application code but the editor’s syntax highlighter, which registered all 190 language grammars it ships with and produced a 1.4 MB chunk on the three heaviest pages; registering only the languages a sales editor plausibly needs removed roughly 290 kB from each, and deferring the editor itself on the AI inbox took that page from 935 kB to 543 kB of first-load JavaScript.

Re-measured on the production build in desktop mode, performance averages **98** against 52, and the login page’s LCP falls from 41.3s to 1.3s with total blocking time at zero. Accessibility rose from 83 to 92 on four distinct faults: two contrast ratios below the 4.5 threshold, an `aria-label` on a bare `div` where ARIA prohibits it, a missing `main` landmark on both the landing page and the authenticated layout, and a viewport meta that disabled pinch-zoom. Best Practices reached 100 once the pages were served from the production server rather than over HTTP on localhost.

Those are asset and bundle sizes measured from the production build. **Table 4 is the pre-remediation measurement and was not re-run**, so the improvement is reported in the quantity that changed rather than asserted as a new score. The remaining bundle weight is dominated by the rich-text editor, which is still loaded before it is displayed.

5.2 Backend performance

We load-tested six interactive read routes — campaigns, meetings, the daily-action summary, outcome metrics, team productivity and outreach settings —

Table 4. Lighthouse category scores (0–100), before → after remediation

Page	Performance	Accessibility	Best Practices	SEO
/ (login)	54 → 97	81 → 100	73 → 100	100
/ai-inboxes	47 → 99	81 → 87	100	100
/outreach	55 → 97	87 → 90	100	100
Average	52 → 98	83 → 92	91 → 100	100

none calling a model. Each level ran three times for forty seconds, median reported; an earlier single run showed lower throughput at twenty-five users than at fifty, which is noise, and repeating removes it.

Two load shapes are reported because they answer different questions (Table 5). With a two-second think time between requests, which is what a person browsing actually does, latency is *flat* from ten users to two hundred: p95 stays at 7–8 ms throughout and throughput rises linearly with the number of users, so the service is nowhere near its limit at two hundred concurrent sessions. With no think time — each thread issuing its next request the instant the last returns, roughly a hundred requests per second per thread — throughput plateaus at 852 requests per second while latency grows in proportion to concurrency. That is the signature of a saturated service queueing rather than one failing: every one of the requests completed, at every level, in both shapes.

Table 5. Interactive routes under load, median of three runs (no AI calls)

Users	requests/s	p50	p95	p99
<i>Two-second think time (browsing users)</i>				
10	4.9	4.2 ms	8.1 ms	9.7 ms
25	12.1	4.0 ms	7.0 ms	8.9 ms
50	24.1	4.0 ms	7.5 ms	9.5 ms
100	48.5	3.9 ms	7.2 ms	9.0 ms
200	96.7	3.7 ms	7.0 ms	9.2 ms
<i>No think time (saturation)</i>				
10	627	11.7 ms	36.1 ms	49.3 ms
25	635	32.0 ms	84.0 ms	125.4 ms
50	721	61.4 ms	123.7 ms	157.9 ms
100	824	105.6 ms	218.2 ms	284.5 ms
200	852	191.0 ms	434.9 ms	566.1 ms

Success rate was 100% at every level in both shapes.

Three limits on what this shows. The routes exclude every path that calls a model — that is the point being supported, since AI work is queued to background workers and the paths a user navigates do not wait on it — but it means these figures describe the interactive tier alone. The test runs against a single local process with the database on the same host, so the numbers characterise the application’s behaviour under concurrency, not a production deployment’s

capacity. And per-request rate limiting, enabled in normal operation, was disabled for the measurement: with it on, an early run had 7,592 of 7,675 requests refused, so the figure produced was the limiter’s throughput rather than the request path’s.

5.3 AI function: reply drafting protocol

We assess reply drafting with rubric-anchored LLM judges in the manner of G-Eval [16]. Because automated judging is biased [14, 15], the protocol is hardened four ways. First, to neutralize self-preference [17], candidate and judge pools are kept disjoint: the three judges come from three vendors, none of which produces a candidate — the candidates are COSMO, Claude Sonnet 5, and Gemini 3.1 Pro (Table 6). Second, since a judge weaker than the models it grades cannot separate their outputs, the panel is drawn from the strongest models outside the candidate families — Nemotron-3-Ultra-550B (NVIDIA), top-ranked on the Artificial Analysis Intelligence Index [19] at the time of writing, Mistral Large, and Qwen 3.6 — with adequacy checked against inter-rater and human agreement. Third, all three outputs go to each judge in one prompt, anonymized as A/B/C with shuffled order, controlling position bias. Fourth, judges run at temperature zero, with an exception on *next-step* for `DO_NOT_CONTACT/OUT_OF_OFFICE` inputs (a respectful close without a sales call-to-action scores highest), and cells diverging by two or more points are aggregated with the median.

The benchmark is designed at 243 inputs (27 seed messages, one per label of the intent taxonomy in Table 2, each augmented into paraphrased variants); results below use the 27-item core set with complete generation from all three candidates, giving 81 judgements per model per criterion (27×3 judges), while scoring the full set continues. COSMO outputs come *end-to-end from the running system* — production endpoints with retrieval over the company knowledge base — while baselines get the identical email, knowledge, and length/no-signature instruction.

5.4 Reply generation results

Table 6 reports per-criterion means, collected under the protocol above. All three candidates score at or near the ceiling on tone and groundedness: given the same knowledge, each model states the correct price (e.g., COSMO: “...Pricing is \$99/seat/month for the Growth plan, or \$249/seat/month for the Scale plan, which includes SSO and a dedicated customer success manager...”). The differences lie in *conciseness*, where COSMO’s length constraint holds it at 5.00 while Gemini 3.1 Pro runs longer (4.77), and in the *next-step* criterion, where COSMO more consistently closes with a specific action (4.56 against 4.26 and 4.33). The margin over Claude Sonnet 5 is small (4.88 against 4.81) and not statistically distinguishable (paired Wilcoxon on per-judgement averages, $p = 0.09$): the result is genuine parity with the frontier baselines on grounded drafting, obtained from the deployed pipeline rather than a tuned experimental setup.

Table 6. LLM-as-a-Judge Results for Reply Generation (scale [1–5])

Metric	COSMO	Claude Sonnet 5	Gemini 3.1 Pro
Relevance	4.93	4.84	4.99
Tone	4.93	4.98	5.00
Groundedness	5.00	5.00	5.00
Conciseness	5.00	4.95	4.77
Next-step	4.56	4.26	4.33
Average	4.88	4.81	4.82

5.5 Reliability and discussion

To ensure the realistic aspect of COSMO, we discuss the reliability of the evaluation results about reply drafting protocol. In particular, inter-judge agreement (Krippendorff’s alpha [18], ordinal) is 0.34 on reply drafting. On reply drafting, the judges do not agree, and the reason lies in the raw scores: three of five criteria sit at 4 or 5 in every one of 405 cells, so nearly all the variance comes from *next_step* and *conciseness*. Exact agreement is nevertheless 89.0% and 91.1%; cells diverging by two or more points (51 of 405, 28 of 243) were aggregated with the median. Alpha compares observed disagreement against what chance would produce given the marginal distribution, so a ceiling-bound scale with little variance yields a low alpha despite high raw agreement. The 0.34 reads as *this rubric does not discriminate between three strong systems*, not *the judges are unreliable*: the reply ranking is weakly supported.

On the other hand, two raters with B2B sales-development experience independently scored ten sampled emails, 126 paired scores each, blind to model identity with order shuffled independently per email, neither seeing the other’s scores.

Agreement between the two raters is 0.31 on reply drafting, with 81.7% of all cells within one point. On reply drafting, the panel agrees with the raters at only 0.29: a rejection email had one rater score every candidate 1 on *next_step* for attaching a demo link to a decline, while the other scored every candidate 5, reading the same replies as politely closing.

As a result, two claims should be kept apart. The audit supports a structural claim — COSMO is AI-native and the incumbents are not — as strong as the public documentation it rests on. The judge scores support a narrower claim about output quality on a curated, augmented set under a verified judging protocol, speaking to capability under representative inputs, not production-scale performance.

6 Conclusion and Risks

We gave AI-nativeness a testable meaning through six criteria, audited four platforms against them, built COSMO to satisfy all six, and evaluated it across

frontend, backend, and AI quality, reporting where COSMO matches the frontier baselines for retrieval-grounded drafting. Several risks temper the result: the AI benchmark’s 27-item core set is small and English-only, and scoring the full 243-item augmented set is ongoing; LLM judging retains residual biases a disjoint-pool design mitigates but cannot remove [14, 15]; dependence on external model APIs introduces latency and availability risk; and the human-in-the-loop checkpoint that makes COSMO safe also caps how much it can automate. A larger multilingual benchmark and tying scores to realized sales outcomes remain open.

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